

Functional Requirements Document (FRD)

1. **Introduction**

1.1 Purpose

This document outlines the functional specifications for VET-SOURCE, a blockchain-based veterinary insurance claims processing platform. It translates the business requirements into detailed functional requirements to guide development, integration, and deployment.

1.2 Scope

The system will enable:

- Automated insurance claim submissions by veterinary clinics.
- Smart contract-based claim validation and approvals.
- Secure, immutable data storage on a blockchain network.
- Real-time notifications and tracking for stakeholders.
- Integration with existing insurance and veterinary management systems.

1.3 Audience

This document is intended for developers, system architects, testers, and project managers involved in the implementation of VET-SOURCE.

2. **System Overview**

VET-SOURCE leverages blockchain technology and smart contracts to automate claim processing, ensure data integrity, and enhance interoperability between veterinary clinics and insurance providers. The system is designed to be scalable, secure, and compliant with relevant industry regulations.

3. **Functional Requirements**

3.1 User Roles & Access Control

The system shall support role-based access control with the following user roles:

- **Veterinary Clinics:** Submit insurance claims and track their status.
- **Insurance Providers:** Verify and approve/reject claims.
- **Regulatory Authorities:** Audit claim processing for compliance.
- **Pet Owners:** View claim status and historical records.
- **Administrators:** Manage system settings and user permissions.

3.2 Claim Submission & Processing

- The system shall allow clinics to submit claims via a web-based interface or API.
- The system shall support document uploads (e.g., invoices, medical records) in multiple formats (PDF, JPEG, PNG, etc.).
- Each claim shall be assigned a unique transaction ID recorded on the blockchain.
- The system shall validate claims against predefined smart contract rules.
- Approved claims shall trigger automated fund transfers from insurers to clinics.

3.3 Smart Contract Execution

- The system shall execute smart contracts to:
 - Verify claim data against predefined eligibility criteria.
 - Validate the authenticity of submitted documents.
 - Execute automated payments upon claim approval.
- Smart contracts shall be upgradable to accommodate regulatory changes.

3.4 Data Storage & Security

- All claim data shall be stored immutably on the blockchain.
- Sensitive data shall be encrypted using industry-standard encryption protocols.
- The system shall comply with GDPR and other relevant data protection regulations.

3.5 Notifications & Alerts

- Users shall receive real-time notifications via email, SMS, or in-app alerts for claim status updates.
- Insurance providers shall be alerted for claims requiring manual review.
- Regulatory bodies shall receive periodic reports on claim processing activities.

3.6 API & Integration

- The system shall provide REST and GraphQL APIs for third-party integration.
- Supported data exchange formats include JSON, XML, CSV, YAML, and Protobuf.
- The system shall support communication protocols such as HTTP, WebSockets, MQTT, gRPC, and AMQP.

3.7 Middleware & Communication Management

- Middleware shall enable interoperability between Web2 and Web3 systems.
- The system shall support legacy systems through data transformation and backward compatibility.
- Communication management shall include:
 - Load balancing for high traffic handling.
 - Message queues for asynchronous processing.
 - Event-driven architecture for real-time interactions.

3.8 Reporting & Auditing

- The system shall generate automated reports on claim approvals, rejections, and processing times.
- Regulatory bodies shall have access to an immutable audit log.
- Insurers shall have dashboards displaying real-time claim analytics.

4. **Non-Functional Requirements**

4.1 Performance & Scalability

- The system shall process at least 10,000 transactions per hour without performance degradation.
- The blockchain infrastructure shall support horizontal scaling.

4.2 Security & Compliance

- Multi-factor authentication (MFA) shall be implemented for user access.
- All transactions shall be digitally signed to ensure authenticity.
- The system shall maintain an uptime of at least 99.9%.

4.3 Usability & Accessibility

- The user interface shall follow WCAG 2.1 accessibility standards.
- The system shall support multiple languages for global adoption.

5. **Roadmap**

5.1 MVP (Cloud-Based Deployment)

- Initial launch with cloud-based infrastructure.
- Core functionalities: claim submission, smart contract validation, and notifications.

5.2 Piloting & Testing

- Conduct pilot programs with select veterinary clinics and insurers.
- Gather feedback for feature refinements and compliance adjustments.

5.3 Web2 & Web3 Integration

- Enable seamless connection between traditional Web2 insurance platforms and blockchain-based Web3 infrastructure.
- Implement decentralized identity verification and token-based incentives.

6. **Success Criteria**

- 50% reduction in claim processing time.
- 90% adoption rate among targeted veterinary clinics within the first 12 months.
- Zero reported cases of fraudulent claims due to blockchain verification.

7. **Next Steps**

- Conduct stakeholder review sessions.
- Develop a prototype based on FRD specifications.
- Perform security testing and compliance validation.

This FRD provides a detailed functional roadmap for VET-SOURCE. Let me know if any refinements are needed!